Another Baby Boom? How Same-Sex Marriage and the Affordable Care Act Increased Births in the US

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Abstract

Assistive Reproductive Treatment (ART) clinics use various procedures to help patients facing infertility issues and same-sex couples have children. Using CDC Data, I find that same-sex marriage legalization boosted frozen donated births by 6-10 births per clinic. Due to the improving health profile of individuals receiving ART treatments, this impact is likely driven by same-sex couples' uptake of ART. Additionally, these increases were distinct from the increased usage of ART due to the Affordable Care Act. However, subsequently, these effects diminish at the clinic level due to an increase in the number of clinics and an exhaustion of the demand.

1 Introduction

Since the birth of Louise Brown, the first baby born as a result of in-vitro fertilization (IVF), IVF has become a cornerstone of the fertility industry. Over the past few decades, the use of IVF has expanded rapidly, which now represents over 99% of Assisted Reproductive Technology (ART) procedures in the United States (US Dept of Health and Human Services, 2024). ART is a family of procedures with the goal to help those with a uterus conceive. This is of particular importance, since ART is necessary for the percentage of the population who are unable to conceive due to various reasons related to infertility. In light of declining U.S. fertility rates(Dilmaghani et al., 2024), the importance of ART in supporting population growth and addressing demographic challenges cannot be understated. Therefore, I investigate whether same-sex individuals, through Same-Sex Marriage legalization, impacted the number of births which occurred through ART procedures.

Same-Sex Marriage (SSM) legalization had a staggered state-by-state rollout initially, but was nationally legalized through the Supreme Court's judicial ruling in 2015(Wolf,R., 2015). This policy enabled same-sex couples to formalize their unions, which could have implications for both their financial capabilities (through their partner's insurance and/or tax benefits) and their desires to pursue parenthood (the ability to have both parents be listed on the child's birth certificate). Cisgender same-sex couples are unable to conceive on their own. Therefore, same-sex couples can use ART clinics to receive reproductive material donated by individuals who are of a different sex. This material is then implanted into one of the parents, or a surrogate[1].

In order to evaluate the policy's impact, I used previously unlinked yearly data obtained from ART clinics' mandatory reporting to the Centers for Disease Control (CDC) from 2004 to 2016. This data is a near population of clinics in the US with information on types of procedures used, location of clinics, transfers[2], and births which occurred as a result of those

[1]Surrogates are individuals who are often paid to carry another couple's child. Same-sex men (Who are Cisgender) must participate in surrogacy to conceive.

[2]A transfer is the procedure of implanting a number of embryos into a uterus.

transfers. After linking the clinics across years, I identified which clinics are located in states where SSM had been legalized.

Relating to general SSM papers, most find that SSM led to significant tax breaks and savings through using their partner's health insurance (Downing and Cha, 2020; Friedberg and Isaac, 2024; Piano, Behr and West, 2023). Papers also find increases in labor supply, large financial investments, and increases in employer-sponsored health insurance coverage (Hansen, Martell and Roncolato, 2019; Downing and Cha, 2020). My paper is the first one to investigate how same-sex marriage affected same-sex family formation through a lens other than adoption. Further literature for the intersection of fertility and SSM legalization is discussed in more detail in Section 3.1.

For my dataset, I quantify the effect of SSM legalization using a difference-in-difference approach. I consider a state treated if SSM is legalized and is not later made illegal. Therefore, the control states are those in which SSM is not permanently legal. This means that by 2015, due to the supreme court decision, all states instantaneously become treated at this time. Due to the differential timing of the legalization of SSM, I employ a staggered difference-in-difference model to examine the effects of this policy on births which occurred from procedures in ART clinics.

I find evidence that after SSM legalization, ART clinics saw an increase in thawed donor births of between 6-10 births per clinic over 4 years. This is likely due to thawed procedures being cheaper and donated material being necessary for same-sex couples to give birth to a child. This gives evidence that the increases in births from this legalization are from same-sex individuals. This effect, however, appears transient, with demand for ART treatments returning to pre-policy levels after approximately two years.

One concern occurs with this method of testing is that the Affordable Care Act (ACA) is passed at a very similar time and affected ART clinics through insurance mandates. These insurance mandates required medicaid to cover a certain monetary amount of ART procedures for couples. When I apply this model using the timing of the ACA, the number of both types of births increases.

As a result, I develop a test to examine whether the results from the SSM legalization study were influenced by the ACA. This is done by including the type of transfer for each birth method as a control. The purpose is to determine whether births would still increase after controlling for the number of transfers at that individual clinic. These models only show increases in thawed donated births for SSM legalization, and are flat for the ACA. This strongly suggests that the impact we are seeing from SSM legalization is directly from same-sex couples. Since a transfer is necessary for a birth to occur, if the rate of successful transfers increases, this implies that the population of individuals receiving ART procedures is healthier. There are multiple papers which show positive birth outcomes from ART are more likely if the patient is healthier or does not have fertility issues when compared to individuals who were less healthy or had various fertility issues (Gaskins et al., 2023; Libby et al., 2021; Stern et al., 2015; Declercq et al., 2015; Liang et al., 2022). This evidentiates that a more healthy group of individuals are giving birth using the thawed, donated methods in relation to SSM legalization. Using the same method of adding in transfers as a control and using the ACA's timing, the impact is zero. This implies that increases in insurance mandate coverage does not change the health profile of those receiving ART treatments. This is most likely attributable to same-sex couples deciding to use ART birth methods more often, as same-sex individuals' reasons for using these treatments are not solely related to medical infertility.

I also run the original test using only states which had insurance mandates for ART[3]. This shows nearly the same estimate results as the model with all states, demonstrating that this impact is not specifically driven by states with insurance mandates.

After this, I exploit the timing of SSM legalization again to determine whether the number of clinics in a city changes after each policy. From this, I see that two years after SSM there was a one year increase of about 0.3 clinics per city. This may have contributed

[3]Due to the timing of SSM legalization, there are not enough states to test this for only states without insurance mandates.

to the perceived exhaustion of demand, since new clinics opened up to accommodate the new demand, leading to the per-clinic numbers returning to zero. Unfortunately, robustness checks for the ACA also show an increase in the number of clinics, and it is impossible to separate the effects of these policies with this data.

The rest of this paper is organized as follows. Section 2 is split into two subsections and discusses the backgrounds of these two policies (SSM legalization in Section 2.1, and the ACA in Section 2.2). Section 3 has a more in depth literature review of the two policies (SSM legalization in Section 3.1, and the ACA in Section 3.2). In Section 4, I describe the data I use in this study, and Section 5 explains how I use this data to form my research design and estimation strategy. Section 6 demonstrates the results of this dataset, and Section 7 draws conclusions and reviews my paper's contributions.

2 Background

2.1 Same-Sex Marriage Legalization

Same-Sex Marriage (SSM) legalization is defined as the passage of laws, or constitutional interpretation, that require marriages and marriage licenses between individuals of the same-sex to be recognized as the same as those of opposite-sex couples. As a result, a large number of same-sex couples decided to get married quickly after it was legalized. This led to changes in insurance coverage for both spouses, allowing individuals to seek insurances which covered IVF. I use the timing of SSM legalization to test whether the states that have legalized SSM experienced an increased use of IVF.

It should be noted for estimation purposes that SSM was legalized via three methods: (1) legalization at the legislative level, (2) judicial intervention in which it is legalized by a judge's ruling, and (3) by referendum, in which citizens voted directly on the change in legal status. On top of this, SSM legalization has a staggered timing effect, where some states began legalizing in 2004 while others legalized later. By 2015 a Supreme Court decision forced all states that had not yet done so to legalize SSM immediately (see table 2). This means the years used for this analysis are from 2003-2014, since by 2015 all states are considered treated. Below is Table 1 which shows the legalization timing by state, and includes an overlay that signifies which states had IVF coverage mandated in their state. The SSM legalization method for each state is also given.

Table 1: Year of Treatment by Same-Sex Marriage and Method of Legalization Passed

Year	Judicial	Legislative	Referendum
2004	Massachusetts*		
2008	Connecticut*		
2009	Iowa	Vermont	
2010		New Hampshire, District of Columbia	
2011		New York*	
2012		Washington	Maine
2013	California*, New Jersey*, New Mexico	Delaware, Hawaii*, Minnesota, Rhode Island*	Maryland*
2014	Alaska, Arizona, Colorado, Indiana, Montana*, Nevada, North Carolina, Oklahoma, Oregon, Pennsylvania, South Carolina, Utah, Virginia, West Virginia*, Wisconsin, Wyoming	Idaho, Illinois*	
2015	Alabama, Arkansas*, Florida, Georgia, Kansas, Kentucky, Louisiana*, Michigan, Mississippi, Missouri, Nebraska, North Dakota, Ohio*, South Dakota, Tennessee, Texas*		

Note: Those with * have insurance coverage for IVF that is mandated to some extent in that state. This chart is made using information from (Hansen, Martell and Roncolato, 2019).

2.2 Affordable Care Act

The Affordable Care Act (ACA) was passed in the United States in 2010. The ACA decreases the number of uninsured individuals and implements reforms into the health insurance market. Under this law, individuals who were uninsured due to pre-existing conditions or limited finances are now able to obtain affordable health plans through the health insurance markets. One way this is accomplished is through the expansion of Medicaid to individuals with limited finances. As some states mandated that individuals could now get coverage of IVF through Medicaid as well.

Using variation in timing and across states, I study which states have ART-related insurance mandates which have been expanded (Resolve: The National Infertiity Association, 2023). To account for anticipation affecting the difference-in-difference model, I set the level of legalization to 2010. Unfortunately, a Supreme Court challenge to the ACA led to it being blocked until 2012, thus full access to many of the benefits from the ACA did not roll out until 2014. Due to this delay in rollouts some states do not implement these insurance changes until approximately two or four years after it's initial passage in 2010.

I also use the data six years before and after the legalization, so I only examine 2004-2016. In this time frame, no individual states changed their insurance mandate laws on IVF through legislation other than through the ACA. However, in 2017 the CDC combined all births into one variable. This was likely in preparation for how they presented the data years later, when they made the data more anonymous for clinics which had fewer than five of any type of procedure or outcome. As a result, data after 2016 is not available.

Below, Table 2 shows the states which were impacted by the ACA. In this chart the treated states are the states which had mandated coverage, while untreated states are those which did not have mandated insurance coverage. In this paper, a state becomes treated if the state-mandated coverage of IVF was expanded through the ACA.

Table 2: Affordable Care Act Treated vs. Untreated

Treated in 2010	Arkansas, California, Connecticut, Hawaii, Illinois, Louisiana, Maryland, Massachusetts, Montana, New Jersey, New York, Ohio, Rhode Island, Texas, West Virginia
Never Treated	Alabama, Alaska, Arizona, Colorado, Delaware, Florida, Georgia, Idaho, Indiana, Iowa, Kansas, Kentucky, Maine, Michigan, Minnesota, Mississippi, Missouri, Nebraska, Nevada, New Hampshire, New Mexico, North Carolina, North Dakota, Oklahoma, Oregon, Pennsylvania, South Carolina, South Dakota, Tennessee, Utah, Vermont, Virginia, Washington, Wisconsin, Wyoming

3 Literature Review

3.1 Same-Sex Marriage (SSM) Legalization

Overall the literature that examines SSM legalization in relation to ART and child adoption of LGBT+ individuals is scarce.

Most previous papers which looked into SSM legalization examined insurance access through their partners' insurance. Other papers studied birth rates of same-sex couples after SSM legalization. There is evidence that both the number of children in same-sex households did not change after SSM legalization (Hansen, Martell and Roncolato, 2019) and that there is a downward trend in the number of adoptive same-sex lesbian households after SSM legalization, though these results may violate pretrends (Bielsa, 2024). These papers show SSM legalization may potentially have increased ART births after legalization, since ART and adoption are the only way that same-sex couples can have children.

Several have looked into whether same-sex individuals have better health outcomes in comparison to straight women with infertility issues using ART. Swedish data shows that same-sex lesbian couples had similar birth outcomes to heterosexual individuals not using ART and better outcomes than heterosexuals using ART (Goisis, Cederstrom and Martikainen, 2023). When comparing shared motherhood IVF and artificial insemination with donor sperm, pregnancy rate was higher, but health outcomes were similar (Matorras et al., 2023). These papers show that outcomes of ART are more favorable for lesbian women, primarily because they generally do not have issues with fertility.

3.2 Affordable Care Act (ACA)

The main impact of the ACA is that it increased insurance coverage, which in turn increased the usage of treatments when mandates required insurance companies to cover these treatments. In this paper, I study the general impact of insurance mandates on IVF. Many papers find that insurance mandates such as these can cause women to delay marriage and child birth (Abramowitz, 2013, 2016; Machado and Sanz-de Galdeano, 2015). These declines are most substantial among younger cohorts of women, with a decline in marriage and an increase in tax filings by individuals in these groups (Heim, Lurie and Simon, 2017).

Other papers have evaluated whether the ACA affected ART procedures usage in the states with mandated insurance coverage. One such paper showed that an IVF procedure called intracytoplasmic sperm injection, used in fresh non-donated conceptions, had lower rates of use in states with IVF insurance mandates between 2000 and 2015 (Dieke et al., 2018). Mandates in New Jersey and Connecticut showed greater usage of ART without a significant change in birth outcomes (Crawford et al., 2016). States with mandated insurance coverage also experienced lower rates of discontinuation in treatments after unsuccessful treatments (Lee et al., 2022). Insurance-mandated states also had higher rates of ART use, and the lack of infertility insurance mandates were associated with adverse perinatal outcomes (Boulet et al., 2015). There is also significant debate on whether the ACA, through expansion of Medicaid, had an impact on these medical outcomes (Devine, Stillman and Decherney, 2014).

4 Data

My data comes from the CDC's yearly reporting from ART clinics through their National ART Surveillance System as (Centers for Disease Control and Prevention, US Dept of Health and Human Services, 2023). This is a legally mandated program, which requires clinics that do ART to report their results and procedures to the government. This dataset includes 91.4% of clinics in the US and approximately 98% of all ART cycles[4] performed in the US, and reporting is legally mandatory for all clinics. This data was hand-linked at a clinic-level, matching by name (and previous names), city, and medical director, and includes approximately 750 clinics.

There are several limitations with this data. The first limitation is that, after 2016, the

[4] A cycle is the procedure for harvesting eggs to be implanted with sperm to become embryos.

data values were censored for clinics which have fewer than 5 births per age group, to protect individuals from being identified.

Additionally, there are reporting issues with fresh, non-donated births. Transfers for this procedure were not reported until 2011, with only cycles related to this procedure recorded before then. This could be due to the CDC not requesting the information or poor reporting practices. As a result, clinics might deprioritize outcomes from this procedure. Additionally, testing indicates significant pretrend issues for this birth method. Therefore, models using these outcomes are only included in the Appendix to avoid using compromised data.

Finally, Table 3 below lists summary statistics of the data used in this analysis:

Table 3: Summary Statistics for IVF Clinics

Clinic Information	Average	Standard Deviation
Frozen Donor Transfers	15.400	30.215
Frozen Non-Donor Transfers	75.990	140.918
Fresh Donor Transfers	20.421	39.630
Fresh Non-Donor Cycles	216.537	328.080
Frozen Donor Births	5.459	12.356
Frozen Non-Donor Births	28.855	66.528
Fresh Donor Births	11.125	21.621
Fresh Non-Donor Births	62.115	94.585
Allow Single Women	0.938	0.245
Have Accreditation	0.920	0.271
Accreditation Pending	0.023	0.151
Use Donor Embryos	0.679	0.467
Use Donor Eggs	0.927	0.259
Allow Surrogates	0.833	0.373
Follow ACA Insurance Mandates	0.524	0.500
Clinics Total (2003-2016)	5929	

From most to the least commonly used type of IVF, I have: (1) Fresh non-donated (2) Thawed non-donated (3) Fresh donated, and (4) Thawed donated. A confounding variable may be that not all clinics have cryopreservation available.

5 Empirical Strategy

I use a difference-in-difference approach in order to identify the impact of Same-Sex Marriage (SSM) legalization on Assisted Reproductive Technology (ART) clinics in the US. Below I use the linked ART-clinic data, discussed in Section 4, to create the primary estimating equation. I use "not yet" and "never treated" clinics together as my control group.

5.1 Primary Estimating Equation

This model uses the differentiation by state of the timing of the legalization of same-sex marriage shown above in Table 1:

$$Births_{i,s(i),t(i)} = \sum_{k=-3}^{K} \beta_k (1)\{t - t^*(s) = k\}(SSM_{st}) + \lambda_i + \delta_t + \varepsilon_{it} \tag{1}$$

The independent variables in this model are the births that occur in clinic (i), in state (s), and at time (t). This model's variable of interest is the β at time in relation to the treatment (k) which occurs based on when the state legalized SSM. SSM is a variable that is 1 after SSM legalization has occurred in that state, and is 0 otherwise. This model also has clinic-level (λ) and time-level (δ) fixed effects.

I use the Borusyak, Jaravel, and Spiess (BJS) difference-in-difference with staggered treatment timing's imputation estimator (Borusyak, Jaravel and Speiss, 2024) and the Callaway Sant'Anna (CS) difference-in-difference with staggered treatment timing's estimator (Sant'Anna and Callaway, 2021). I also use and report OLS, even though OLS is biased in a staggered setting.

These group treatment effects are reported using an event study approach. These results compare the clinics in states where SSM has occurred (treatment) and the clinics in states where SSM has not yet occurred (control). Section 6 reports the results for this model in more detail along with other versions of this model being run. The Appendix also estimates this model while replacing the independent variable births with transfers at the clinic level.

5.2 Robustness and the Affordable Care Act

I compare the three birth methods before and after the Affordable Care Act (ACA) is implemented. I use states that have ART coverage insurance mandates which were expanded through passage of the ACA as my treated group, while I use states that never had those mandates as the control group. In this setting, treatment occurs in 2010 or does not occur at all. As a result, this test uses a simple OLS event study and the Borusyak, Jaravel, and Spiess (BJS) difference-in-difference imputation estimator (Borusyak, Jaravel and Speiss, 2024). Both methods are used, as BJS is consistent with OLS and can be more accurate in a non-staggered setting. The bounds of this event study model are from 2004 to 2016 due to data restrictions. This model can be written as:

$$Births_{i,s,t} = \sum_{k=-6}^{K} \beta_k (1)\{t - 2010 = k\}(ART_s) + \lambda_i + \delta_t + \varepsilon_{it} \tag{2}$$

The independent variable in this model are the births that occur in clinic (i), in state (s), and at time (t). This model's variable of interest is the β at time in relation to the treatment (k). ART is a binary variable which is only 1 after the ACA is passed in states with an insurance mandate to cover IVF. This model also has clinic-level (λ), and time-level (δ) fixed effects.

This model uses a very simple difference-in-difference framework. The methods to estimate these results uses ordinary least squares (OLS) and the Borusyak, Jaravel, and Spiess (BJS) difference in difference imputation estimator (Borusyak, Jaravel and Speiss, 2024). These methods are both consistent with the standard difference-in-difference methods.

These group treatment effects are reported using an event study approach. These results compare the clinics in states where the ACA has occurred (treatment) and the clinics in states where the policy had not yet occurred (control) in the years after it's passage in 2010. Section 6 reports the results for this model in more detail. The Appendix also estimates this model, while replacing the independent variable births with transfers at the clinic level.

5.3 City-Effects

The final model below uses both legalization methods and is collapsed to the city level:

$$\#ofClinics_{c,s(c),t(c)} = \sum_{k=-K}^{K} \beta_k(1)\{t - t^*(s) = k\}(Legal_{st}) + \lambda_c + \delta_t + \varepsilon_{ct} \quad (3)$$

In this case, #ofClinics refers to the number of clinics in city (c), in state (s), and at time (t). Legal refers to either SSM legalization, or the passage of the ACA, depending on the policy of concern. Legal is 1 if the policy has occurred and 0 if it has yet to occur. This model is designed to estimate how the separate impact of both the ACA and SSM legalization changes the number of clinics in a city. This model also has clinic-level (λ) and time-level (δ) fixed effects.

This model uses both policies in the ways mentioned in the previous two sections. This means that when Legal is for the ACA, I report BJS and OLS, and when Legal is for SSM legalization, I use BJS, CS, and OLS. Section 6 also reports the results for this model.

5.4 Difference-in-Difference Models

I use these estimators and implement them according to a trio of papers (Roth, 2018; Roth et al., 2023; Roth, 2022). The first paper interprets the new difference-in-difference models when using non-staggered treatment and parallel trends fails. This paper demonstrates the validity of the post-treatment event study estimates holds so long as the parallel trends assumption also holds, yielding results even under heterogeneous treatment effects for a non-staggered treatment effect design (Roth, 2018). Results for the event study regarding the ACA fall under this definition, but for robustness, the classic difference-in-difference model with fixed effects is also run; it shows consistent results. The second paper explains when it is optimal to use the new staggered timing difference-in-difference estimators. In that paper they state that "BJS estimator may be preferable in settings where the outcome is not too serially correlated and the researcher is confident in parallel trends across all periods" (Roth et al., 2023). Most models have pretrends which are very consistent over longer time periods, but some analyses have issues with pretrends in the CS models. This, coupled with a fairly high level of serial correlation, is why both models are reported in the staggered setting, as together they form a set of bounds on the results. This is supported by both models generally having similar results, with the CS model having consistently larger standard errors and being closer to zero. This model having larger bounds, can be explained due to the requirement that pretrends are held approximately, the model not using all available pretrends, and/or the issues with serial correlation. My conclusions use all appropriate models for their results.

I use additional pretesting according to the pretrends package in (Roth, 2022). This uses a power of 80% and the Likelihood Ratio Test and Bayes Test results for each model are reported in the appendix. For the Bayes test, in the presence of a significant pretrend in the model, the smaller the value, the more likely that pretrends hold. If the Likelihood Ratio Test is small, it means the coefficients in the data are more likely under parallel trends than under the hypothesized trend. In this case, both of these coefficients must be small if parallel trends has the potential to be violated. These tests can only be completed for the CS and OLS models, since the method requires a normalization based on a pre-period, which by construction, the BJS model does not have.[5]

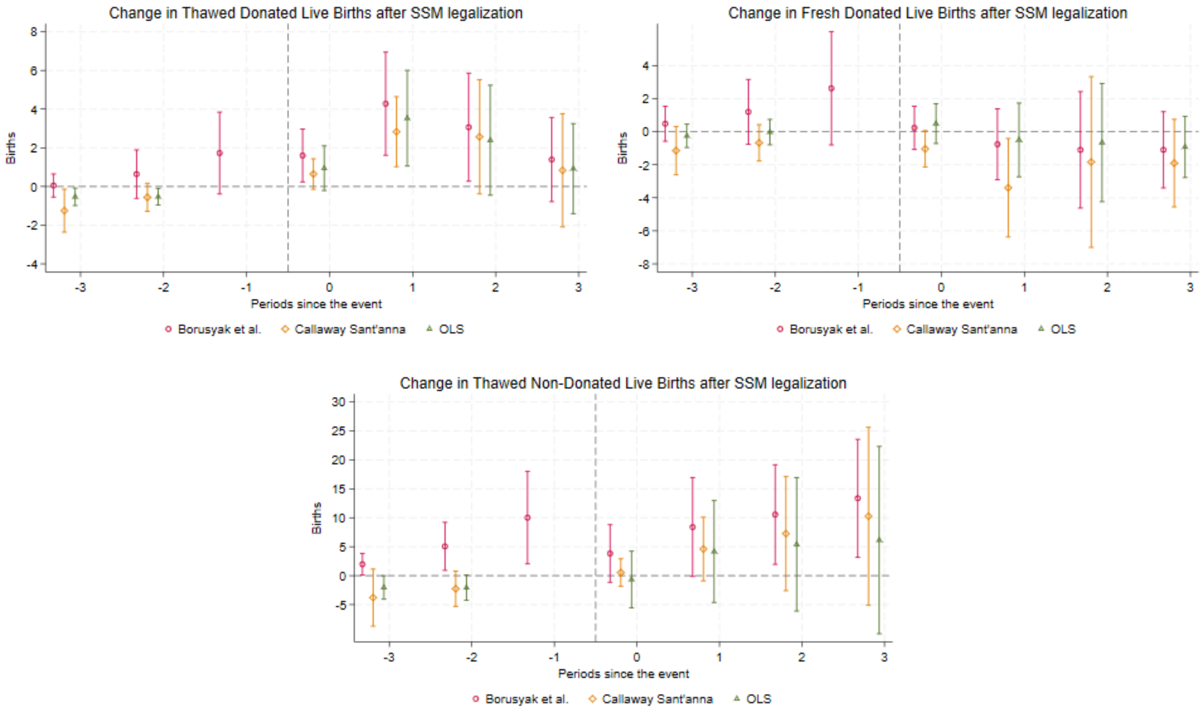
6 Results

6.1 Primary Results

Figure 1 below first examines the results from Equation 1, split by the primary three types of births:

[5]BJS and OLS results will also be reported for all equations, but due to lower sample size, equation 3 results cannot report CS results

Figure 1: Births only increase for thawed donated births after SSM legalization



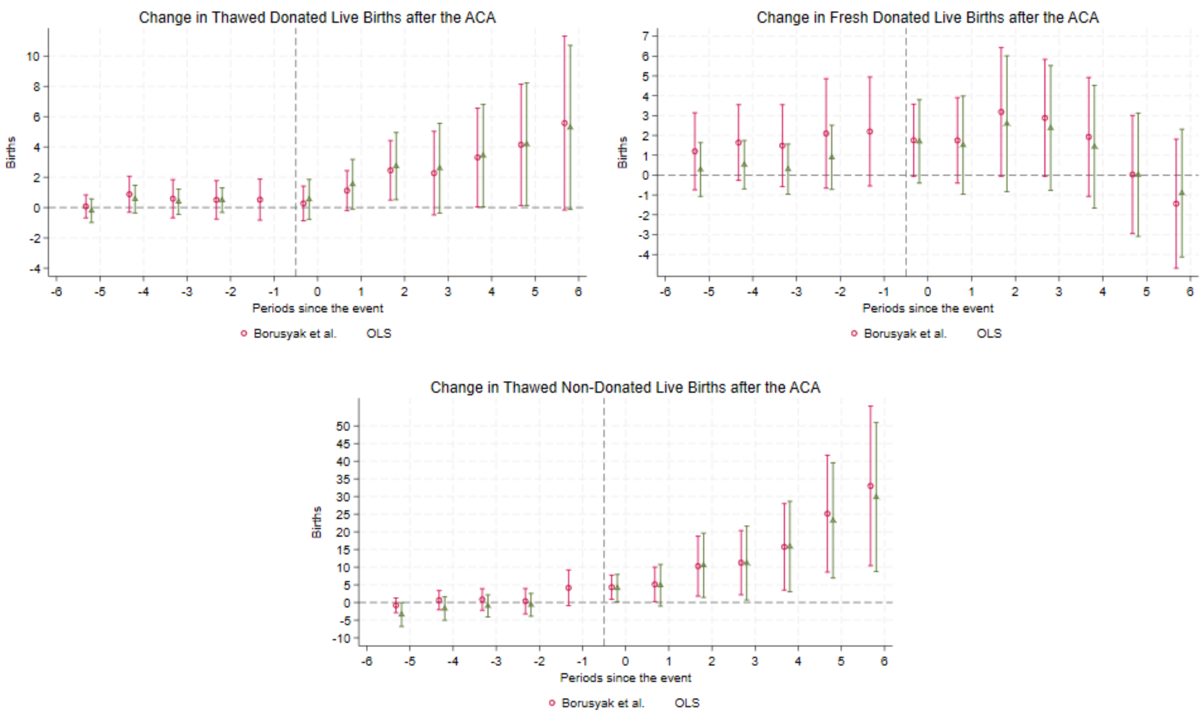
Note: These event study graphs display the results from equation 2 on each different type of births. These charts display the information using a 95% confidence interval.

The first graph in Figure 3 indicates an immediate increase in thawed donor births that lasts about two years and then returns to zero. The overall cumulative increase in these births is between 6.7-10.1 births per clinic across four years. For the model using fresh donor births, these decrease slightly in the year after the event, which could be caused by some individuals switching to frozen donor births. This effect is only statistically significant for the CS model, which also has significant issues with their joint pretrend test, but passes the additional testing for both the Likelihood Ratio, and Bayes tests which are reported in the Appendix. For the third graph, there are some issues with pretrends using BJS. Although this is only the case with thawed non-donor births model, it is accompanied by low values for both the Likelihood Ratio, and Bayes tests reported in the Appendix. These results indicate that SSM caused some changes in the ways ART is used by increasing the number of thawed donor births.

6.2 Affordable Care Act and Transfer Results

With the results from the SSM model showing that births are increasing, I then analyze the ACA model to determine whether the results are similar. The results for this are displayed below in Figure 2 using Equation 2, and are separated by birth type.

Figure 2: Births increase in all three categories two years after the ACA is passed

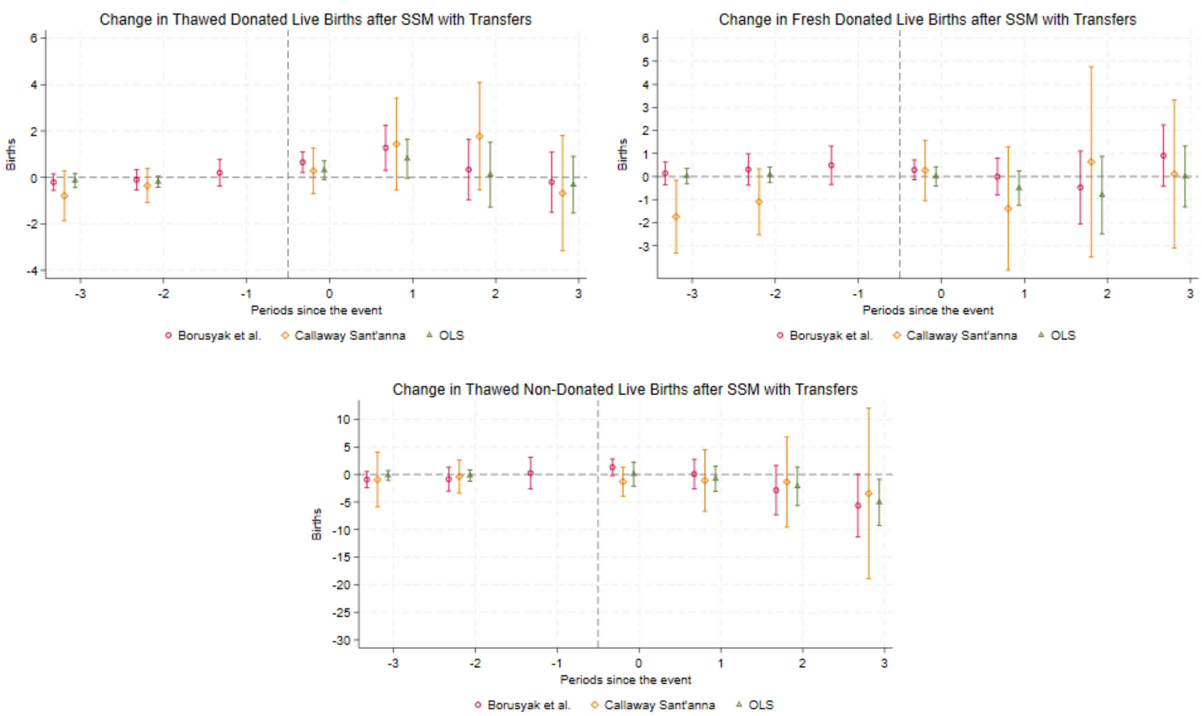


Note: These event study graphs display the results from equation 1 on each different type of births. These charts display the information using a 95% confidence interval.

The first chart in Figure 2 shows that thawed donated births increased approximately two years after the passage of the ACA and had a continual increase in the following years. BJS and OLS also show cumulative increases of 18-20 births per clinic over the seven years. The second chart shows similar results using fresh donated live births. This chart displays an increase two years after the implementation of the ACA, then a downward trend four years after the legalization, which seems to be a return to zero. The third chart, focusing on thawed non-donated live births, reveals an immediate increase in births after the ACA, with an increase that seems to get larger after each additional year. The BJS and OLS also show a cumulative increase, this time of about 96-99 births per clinic across the seven years. These findings point to a significant increase two years after the passage of the ACA for all three listed types of births, with sustained growth in births that are from thawed embryos. This is likely due to thawed treatments both being cheaper than fresh IVF treatments and being much less stressful for those undergoing ART cycles. Overall, these findings indicate that the ACA's insurance coverage had increased the number of births facilitated by ART.

Unfortunately, due to similar timing, this does not prove that the effects of the policies are separate. As a result, I reuse Equation 1 while including transfers as a control variable. The results for Equation 1 with transfers is displayed below in Figure 3 and separated by birth type.

Figure 3: Thawed donated live births is the only one to show an increase that is statistically significant after SSM with transfers



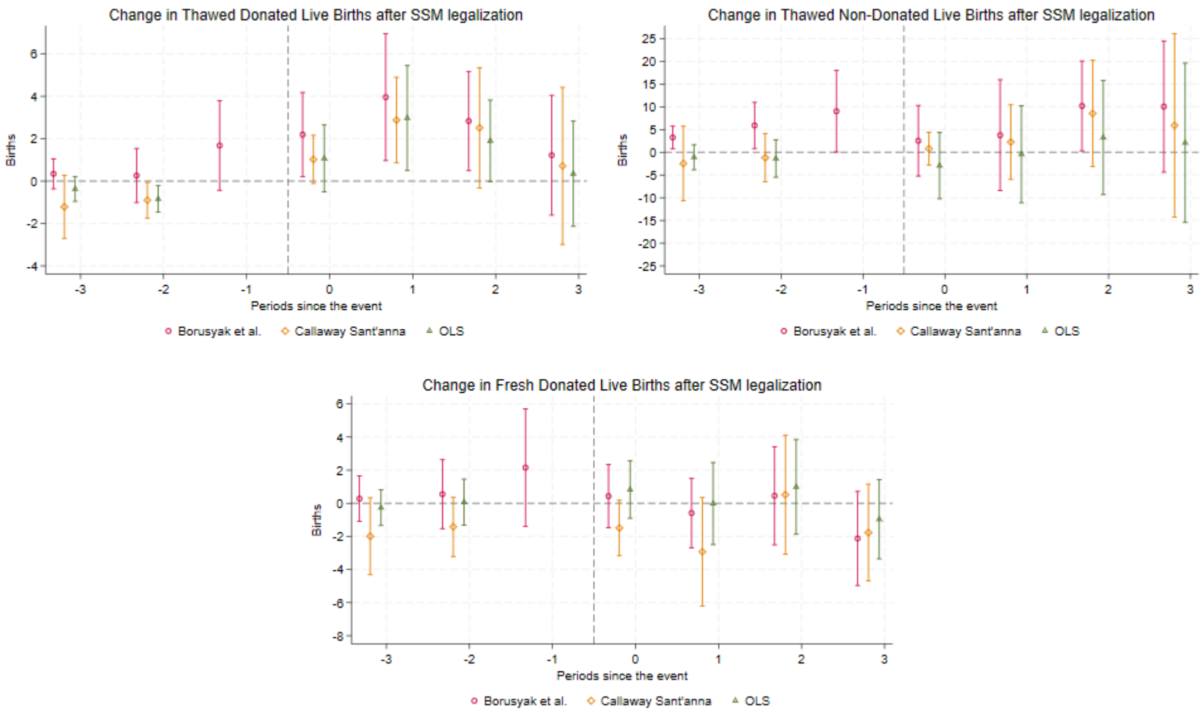
Note: These event study graphs display the results from equation 2 on each different type of births. These graphs also include the control of transfers for their respective type of birth/procedure. These charts display the information using a 95% confidence interval.

As demonstrated above, this model has none of the same pretrends issues from Figure 1. Looking at the thawed donor births model, there is still a significant impact for the year of the policy change and year after in the BJS model. These models do not seem to have significant cumulative effects, due to the return to zero and the large confidence intervals for the T+2 and T+3 periods. At the same time, fresh donated births no longer have statistically significant decreases. It is also apparent that thawed non-donated births no longer have any impact on births. This evidentiates that same-sex couples are the ones that are influencing the increase in thawed donor births.

The split from the ACA is further shown in the appendix, where Equation 2 is similarly run with transfers as a control, resulting no statistically significant outcomes for any type of births. This is because outcomes for births in relation to transfers should not be influenced by insurance coverage for infertile individuals. Insurance coverage and SSM are incredibly unlikely to affect the fertility of individuals using ART. Therefore, this increase indicates that same-sex couples are the drivers in the increase of demand because their ART usage is not based on medical infertility. Instead, it is driven by healthier individuals joining the population of individuals receiving ART treatments and making the average member of the population healthier. This evidentiates that the impact of SSM legalization caused an increase in births in the US that is separate from the ACA and that an increase in birth outcomes for thawed donor births specifically occurred as a result of same-sex individuals.

Lastly, I test whether the impact of SSM is greater in states that had insurance mandates, in order to provide additional proof that these policies are distinct. This is done below in Figure 4, using Equation 1. The data is limited to only clinics that had insurance mandates, and is separated by birth type.

Figure 4: Thawed donor live births increase even more in states with mandated insurance coverage after SSM



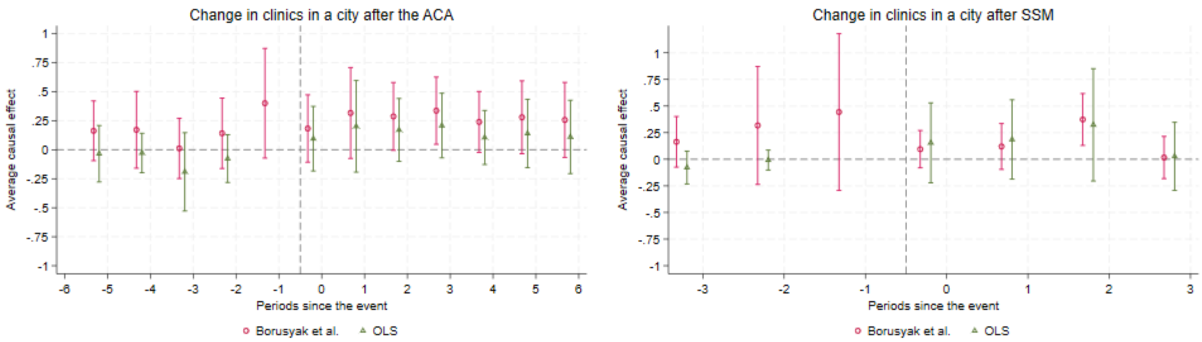
Note: These event study graphs display the results from equation 2 on each type of birth, except it has been limited to only states with insurance coverage that is mandated. These charts display the information using a 95% confidence interval.

Here, the data is still consistent with the previous results, where thawed donated births increase, thawed non-donated births do not change, and fresh donated births either decrease or do not change. This result suggests that states which had insurance mandates experienced very similar effects to those without these mandates, and that the effects of these policies were distinct.

6.3 City-Level Results

The final analysis is to look into whether either of the policies impacted the number of clinics from the supply side, and if there is a potential reason for the exhaustion of demand in Figure 1. Using Equation 3, I create Figure 5 below which is separated by policy.

Figure 5: The number of clinics in a city increase two years after each policy



Note: These event study graphs display the results from equation 3 on each type of legalization. These charts display the information using a 95% confidence interval.

These event studies show the change in the number of clinics within a city after the ACA and SSM legalization. There appears to be an increase in the number of clinics two years later for the SSM model. The ACA model shows an increase in the number of clinics a year after the policy goes into effect, though many of the individual timings are not significant other than three years after the policy. This effect is persistent for the full six years. These results show that the two policies likely influenced an increase in the number of clinics in a city to keep up with the demand of more procedures and births.

7 Conclusions

ART and IVF are crucial for increasing birth rates, and specifically allow older women, infertile women, and same-sex couples to have more children. In this paper, I found that, following Same-Sex Marriage (SSM) legalization, thawed donated live births increased by between 6-10 births per clinic between the year of legalization and three years later. However, this increase only lasted a couple of years at the clinic level.

After this, I needed to separate the effect from SSM from the ACA, which occurred at a similar time, and similarly had increases to thawed births after the policy was codified. Using transfers as a control variable, I found that the increase in SSM models was specifically driven by same-sex couple's usage of ART due to same-sex couple's higher average health. Additionally, I was also able to produce some evidence that clinics did not have a substantially different impact whether their states had insurance mandates or not.

Finally, I was able to show that, after both policies, there were increases in the number of clinics in that state. This however cannot be separated by policy, but these increases seems to be at least partially due to an increase in demand by same-sex couples immediately after the legalization of SSM. The increase in clinics also gives an explaination as to why the results after SSM legalization returns to zero. The other option is that after SSM legalization, the demand from same-sex couples was satisfied.

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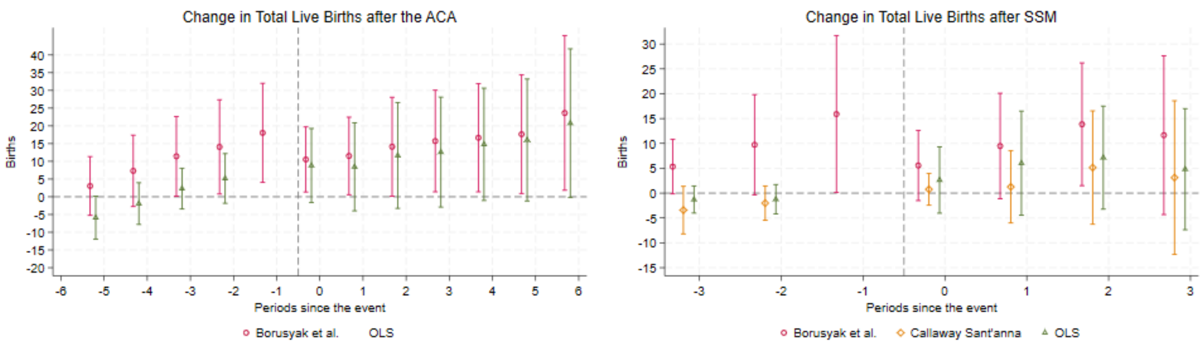
A Appendix

Here I have the charts for the results listed above and extra robustness checks:

Figure 6: Total births increase for both policies with pretrends issues



Note: These event study graphs display the results from Equation 1 on total births without transfers. These charts display the information using a 95% confidence interval.

Table 4: Results For Total Births after the ACA

	(1) BJS Total Births	(3) OLS Total Births
T+0	10.53**	8.809*
	(4.685)	(5.332)
T+1	11.52**	8.444
	(5.581)	(6.334)
T+2	14.13**	11.65
	(7.113)	(7.616)
T+3	15.72**	12.60
	(7.309)	(7.912)
T+4	16.65**	14.80*
	(7.781)	(8.084)
T+5	17.64**	16.02*
	(8.528)	(8.774)
T+6	23.63**	20.75*
	(11.12)	(10.71)
T-1	18.03**	
	(7.133)	
T-2	14.07**	5.170
	(6.785)	(3.575)
T-3	11.42**	2.300
	(5.729)	(2.934)
T-4	7.312	-1.926
	(5.136)	(3.000)
T-5	3.042	-5.885*
	(4.227)	(3.082)
T-6	2.823	-5.449
	(3.217)	(3.731)
N	5773	6052
Joint Pretrends Tests	0.169	-
Cumulative Effect	109.8275**	93.06861*
	(48.56958)	(51.61761)
Likelihood Ratio Test	-	0.004
Bayes Test	-	0.243

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 1 on all births without transfers, and are the tabular results for Figure 6.

Table 5: Results For Total Births after SSM

	(1) BJS	(2) CS	(3) OLS
	totbirths	totbirths	totbirths
T+0	5.321	1.644	2.718
	(3.796)	(2.176)	(3.763)
T+1	7.051	0.247	5.494
	(5.955)	(5.756)	(5.866)
T+2	10.01	4.020	6.383
	(8.761)	(7.274)	(7.635)
T+3	9.823	6.588	2.932
	(12.74)	(11.55)	(10.18)
T-1	15.90*		
	(9.149)		
T-2	10.000*	-0.104	-0.960
	(5.847)	(1.859)	(1.498)
T-3	5.217*	-0.551	-1.284
	(3.158)	(2.534)	(1.472)
N	4872	-	4923
Joint Pretrends Tests	0.379	0.613	-
Cumulative Effect	40.554**	10.275	20.619
	(17.947)	(16.260)	(16.378)
Likelihood Ratio Test	-	0	0.141
Bayes Test	-	0	0.218

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 1 on all births without transfers, and are the tabular results for Figure 6.

Table 6: Results for Thawed Donor Births after SSM legalization

	(1) BJS ThwDnrLvBirths	(2) CS ThwDnrLvBirths	(3) OLS ThwDnrLvBirths
T+0	1.600**	0.644	0.945
	(0.695)	(0.402)	(0.589)
T+1	4.279***	2.833***	3.531***
	(1.359)	(0.922)	(1.262)
T+2	3.065**	2.569*	2.391
	(1.426)	(1.506)	(1.451)
T+3	1.393	0.837	0.923
	(1.110)	(1.493)	(1.187)
T-1	1.728		
	(1.079)		
T-2	0.638	-0.560	-0.528**
	(0.638)	(0.372)	(0.220)
T-3	0.0498	-1.249**	-0.533**
	(0.304)	(0.566)	(0.231)
N	5062		5129
Joint Pretrends Tests	0.1491	0.8230	-
Cumulative Effect	10.337***	6.882**	7.790**
	(3.296)	(3.433)	(3.473)
Likelihood Ratio Test	0	0	13.853
Bayes Test	0	0	0.220

Standard errors in parentheses
* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These are the results for Equation 1 without transfers, and are the tabular results for Figure 1.

Table 7: Results for Thawed Non-Donor Births after SSM legalization

	(1) BJS ThwNDLvBirths	(2) CS ThwNDLvBirths	(3) OLS ThwNDLvBirths
T+0	3.829	0.546	-0.662
	(2.536)	(1.206)	(2.495)
T+1	8.389*	4.598	4.170
	(4.351)	(2.802)	(4.488)
T+2	10.55**	7.257	5.405
	(4.387)	(5.026)	(5.876)
T+3	13.37**	10.26	6.138
	(5.193)	(7.829)	(8.250)
T-1	10.01**		
	(4.072)		
T-2	5.061**	-2.257	-2.066*
	(2.117)	(1.556)	(1.115)
T-3	1.972**	-3.792	-2.039**
	(0.957)	(2.516)	(1.035)
N	5062		5129
Joint Pretrends Tests	0.1796	0.0780	-
Cumulative Effect	36.141***	22.664	15.051
	(12.574)	(14.827)	(18.243)
Likelihood Ratio Test	0	0	1.535
Bayes Test	0	0	.218

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These are the results for Equation 1 without transfers, and are the tabular results for Figure 2.

Table 8: Results for Fresh Donor Births after SSM legalization

	(1) BJS	(2) CS	(3) OLS
	FshDnrBirths	FshDnrBirths	FshDnrBirths
T+0	0.234	-1.041*	0.488
	(0.669)	(0.561)	(0.615)
T+1	-0.760	-3.392**	-0.499
	(1.095)	(1.521)	(1.138)
T+2	-1.100	-1.835	-0.662
	(1.797)	(2.636)	(1.824)
T+3	-1.097	-1.905	-0.921
	(1.177)	(1.352)	(0.946)
T-1	2.624		
	(1.747)		
T-2	1.201	-0.675	-0.0259
	(1.000)	(0.561)	(0.391)
T-3	0.483	-1.147	-0.252
	(0.540)	(0.748)	(0.361)
N	5062		5129
Joint Pretrends Tests	0.5437	0.0058	-
Cumulative Effect	-2.722	-8.173*	-1.594
	(3.430)	(4.730)	(3.556)
Likelihood Ratio Test	0	0	0.026
Bayes Test	0	0	0.218

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These are the results for Equation 1 without transfers, and are the tabular results for Figure 2.

Table 9: Results for Fresh Non-Donor Births after SSM legalization

	(1) BJS FshNDLvBirths	(2) CS FshNDLvBirths	(3) OLS FshNDLvBirths
T+0	-0.101	0.609	1.867
	(1.843)	(1.261)	(1.812)
T+1	-2.427	-2.784	-1.172
	(2.550)	(2.828)	(2.462)
T+2	1.331	-2.850	0.00410
	(4.480)	(5.836)	(4.281)
T+3	-2.006	-6.072	-1.328
	(6.974)	(9.924)	(6.403)
T-1	1.539		
	(4.403)		
T-2	2.837	1.485	1.382
	(2.970)	(1.396)	(1.089)
T-3	2.824	2.771	1.540
	(1.958)	(2.016)	(1.083)
N	5062		5129
Joint Pretrends Tests	0.1373	0.2354	-
Cumulative Effect	-3.203	-11.098	-0.628
	(13.506)	(17.678)	(12.192)
Likelihood Ratio Test	0	0	0.001
Bayes Test	0	0	0.218

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These are the results for Equation 1 without transfers.

Table 10: Results For Thawed Donor Births after the ACA

	(1) BJS ThwDnrLvBirths	(3) OLS ThwDnrLvBirths
T+0	0.274	0.541
	(0.583)	(0.671)
T+1	1.120*	1.539*
	(0.669)	(0.836)
T+2	2.457**	2.732**
	(1.001)	(1.131)
T+3	2.270	2.591*
	(1.407)	(1.510)
T+4	3.308**	3.430**
	(1.661)	(1.724)
T+5	4.143**	4.179**
	(2.042)	(2.061)
T+6	5.571*	5.285*
	(2.926)	(2.762)
T-1	0.521	
	(0.689)	
T-2	0.506	0.488
	(0.646)	(0.413)
T-3	0.581	0.386
	(0.641)	(0.427)
T-4	0.878	0.558
	(0.607)	(0.466)
T-5	0.0779	-0.206
	(0.386)	(0.398)
T-6	-0.236	-0.633
	(0.314)	(0.456)
N	5773	6052
Joint Pretrends Tests	0.835	-
Cumulative Effect	19.14268**	20.29648**
	(9.233787)	(9.758065)
Likelihood Ratio Test	-	0.006
Bayes Test	-	0.245

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 2 without transfers, and are the tabular results for Figure 2.

Table 11: Results For Thawed Non-Donor Births after the ACA

	(1) BJS ThwNDLvBirths	(3) OLS ThwNDLvBirths
T+0	4.329**	4.096**
	(1.751)	(1.965)
T+1	5.130**	4.864
	(2.487)	(3.006)
T+2	10.27**	10.53**
	(4.325)	(4.642)
T+3	11.26**	11.15**
	(4.641)	(5.332)
T+4	15.74**	15.83**
	(6.244)	(6.533)
T+5	25.15***	23.21***
	(8.442)	(8.313)
T+6	32.99***	29.83***
	(11.52)	(10.76)
T-1	4.154	
	(2.580)	
T-2	0.355	-0.668
	(1.830)	(1.650)
T-3	0.853	-0.978
	(1.560)	(1.600)
T-4	0.653	-1.720
	(1.381)	(1.686)
T-5	-0.815	-3.416***
	(1.054)	(1.721)
T-6	-0.357	-3.402*
	(0.765)	(1.892)
N	5773	6052
Joint Pretrends Tests	0.207	-
Cumulative Effect	104.885***	99.506***
	(36.195)	(37.745)
Likelihood Ratio Test	-	0.005
Bayes Test	-	0.230

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 2 without transfers, and are the tabular results for Figure 2.

Table 12: Results For Fresh Donor Births after the ACA

	(1) BJS	(3) OLS
	FshDnrBirths	FshDnrBirths
T+0	1.757*	1.698
	(0.928)	(1.071)
T+1	1.750	1.513
	(1.099)	(1.263)
T+2	3.185*	2.592
	(1.654)	(1.745)
T+3	2.882*	2.371
	(1.505)	(1.605)
T+4	1.926	1.431
	(1.528)	(1.578)
T+5	0.0301	0.0192
	(1.520)	(1.586)
T+6	-1.438	-0.906
	(1.659)	(1.643)
T-1	2.198	
	(1.399)	
T-2	2.101	0.896
	(1.408)	(0.825)
T-3	1.483	0.300
	(1.053)	(0.644)
T-4	1.642*	0.526
	(0.973)	(0.621)
T-5	1.201	0.282
	(0.991)	(0.690)
T-6	0.214	-0.532
	(0.738)	(0.661)
N	5773	6052
Joint Pretrends Tests	0.675	-
Cumulative Effect	10.091	8.718
	(8.049)	(9.050)
Likelihood Ratio Test	-	0.004
Bayes Test	-	0.247

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 2 without transfers, and are the tabular results for Figure 2.

Table 13: Results For Fresh Non-Donor Births after the ACA

	(1) BJS FshNDLvBirths	(3) OLS FshNDLvBirths
T+0	4.166	2.475
	(2.895)	(3.057)
T+1	3.521	0.528
	(3.061)	(3.243)
T+2	-1.787	-4.200
	(3.544)	(3.594)
T+3	-0.691	-3.510
	(3.574)	(3.789)
T+4	-4.327	-5.894
	(3.842)	(3.977)
T+5	-11.68**	-11.39**
	(4.976)	(4.860)
T+6	-13.49**	-13.46**
	(5.711)	(5.302)
T-1	11.16**	
	(4.935)	
T-2	11.11**	4.454*
	(4.612)	(2.421)
T-3	8.506**	2.591
	(4.214)	(2.237)
T-4	4.139	-1.290
	(3.512)	(1.857)
T-5	2.578	-2.545
	(3.011)	(1.949)
T-6	3.202	-0.882
	(2.366)	(2.218)
N	5773	6052
Joint Pretrends Tests	0.055	-
Cumulative Effect	-24.29067	-35.45193
	(21.95437)	(23.24138)
Likelihood Ratio Test	-	0.001
Bayes Test	-	0.246

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 2 without transfers.

Table 14: Results for Thawed Donor Births controlling for Transfers after SSM legalization

	(1) BJS ThwDnrLvBirths	(2) CS ThwDnrLvBirths	(3) OLS ThwDnrLvBirths
T+0	0.652***	0.284	0.307
	(0.227)	(0.503)	(0.204)
T+1	1.274**	1.435	0.811*
	(0.496)	(1.011)	(0.429)
T+2	0.336	1.774	0.117
	(0.662)	(1.180)	(0.713)
T+3	-0.205	-0.680	-0.315
	(0.660)	(1.269)	(0.623)
T-1	0.203		
	(0.295)		
T-2	-0.104	-0.353	-0.188
	(0.223)	(0.374)	(0.122)
T-3	-0.207	-0.792	-0.139
	(0.181)	(0.546)	(0.154)
Transfers	0.380***		0.397***
	(0.0102)		(0.0141)
N	5062		5129
Joint Pretrends Tests	0.227	0.827	-
Cumulative Effect	2.057	2.813	0.920
	1.718	3.057	1.672
Likelihood Ratio Test	0	0	1.908
Bayes Test	0	0	0.221

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These are the results for Equation 1 controlling for the type of transfer used for that birth type, and are the tabular results for Figure 3.

Table 15: Results for Thawed Non-Donor Births controlling for Transfers after SSM legalization

	(1) BJS ThwNDLvBirths	(2) CS ThwNDLvBirths	(3) OLS ThwNDLvBirths
T+0	1.291*	-1.298	0.0682
	(0.766)	(1.347)	(1.112)
T+1	0.0779	-1.076	-0.781
	(1.365)	(2.849)	(1.166)
T+2	-2.847	-1.366	-2.154
	(2.286)	(4.185)	(1.796)
T+3	-5.636*	-3.453	-5.054**
	(2.892)	(7.901)	(2.130)
T-1	0.249		
	(1.471)		
T-2	-0.841	-0.390	-0.190
	(1.105)	(1.531)	(0.525)
T-3	-0.923	-0.923	-0.188
	(0.738)	(2.533)	(0.456)
Transfers	0.457***		0.441***
	(0.0353)		(0.0338)
N	5062		5129
Joint Pretrends Tests	0.0705	0.0787	-
Cumulative Effect	-7.113	-7.193	-7.921
	(6.337)	(14.393)	(4.910)
Likelihood Ratio Test	0	0	0.050
Bayes Test	0	0	0.218

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These are the results for Equation 1 controlling for the type of transfer used for that birth type, and are the tabular results for Figure 3.

Table 16: Results for Fresh Donor Births controlling for Transfers after SSM legalization

	(1) BJS FshDnrBirths	(2) CS FshDnrBirths	(3) OLS FshDnrBirths
T+0	0.289	0.264	0.00696
	(0.219)	(0.668)	(0.213)
T+1	-0.00590	-1.388	-0.506
	(0.408)	(1.364)	(0.381)
T+2	-0.473	0.636	-0.801
	(0.805)	(2.105)	(0.857)
T+3	0.909	0.110	0.00377
	(0.680)	(1.637)	(0.674)
T-1	0.490		
	(0.424)		
T-2	0.306	-1.096	0.0762
	(0.347)	(0.729)	(0.172)
T-3	0.139	-1.740**	0.0176
	(0.254)	(0.805)	(0.169)
Transfers	0.507***		0.499***
	(0.0164)		(0.0189)
N	5062		5129
Joint Pretrends Tests	0.839	0.030	-
Cumulative Effect	0.719	-0.379	-1.296
	(1.777)	(4.079)	(1.783)
Likelihood Ratio Test	0	0	0.005
Bayes Test	0	0	0.219

Standard errors in parentheses
* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These are the results for Equation 1 controlling for the type of transfer used for that birth type, and are the tabular results for Figure 3.

Table 17: Results For Thawed Donor Births after the ACA controlling for Transfers

	(1) BJS ThwDnrLvBirths	(3) OLS ThwDnrLvBirths
T+0	-0.186	-0.163
	(0.305)	(0.280)
T+1	-0.00180	-0.0263
	(0.281)	(0.279)
T+2	-0.0415	0.142
	(0.529)	(0.412)
T+3	0.0266	0.0827
	(0.569)	(0.472)
T+4	-0.134	0.110
	(0.648)	(0.480)
T+5	-0.482	0.0537
	(0.994)	(0.729)
T+6	-0.931	-0.168
	(1.129)	(0.775)
T-1	-0.560	
	(0.445)	
T-2	-0.775**	-0.359
	(0.367)	(0.228)
T-3	-0.235	0.0736
	(0.365)	(0.248)
T-4	-0.0274	0.269
	(0.324)	(0.267)
T-5	-0.690**	-0.423*
	(0.270)	(0.238)
T-6	-0.135	0.0313
	(0.276)	(0.298)
Transfers	0.468***	0.431***
	(0.0395)	(0.0176)
N	5773	6052
Joint Pretrends Tests	0.0196	-
Cumulative Effect	-1.749	0.031
	(3.753)	(2.659)
Likelihood Ratio Test	-	4.753
Bayes Test	-	0.249

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These are the results for Equation 2 controlling for the type of transfer used for that birth type.

Table 18: Results For Thawed Non-Donor Births after the ACA controlling for Transfers

	(1) BJS ThwNDLvBirths	(3) OLS ThwNDLvBirths
T+0	-0.489	-0.386
	(0.796)	(0.712)
T+1	-1.045	-0.715
	(1.233)	(1.026)
T+2	0.606	0.430
	(1.482)	(1.221)
T+3	0.452	0.421
	(1.663)	(1.416)
T+4	-1.289	-0.574
	(2.256)	(1.703)
T+5	-2.415	-1.477
	(2.918)	(1.934)
T+6	-0.606	0.00731
	(3.552)	(2.229)
T-1	-2.279*	
	(1.177)	
T-2	-2.876**	-1.237*
	(1.126)	(0.716)
T-3	-1.945*	-0.545
	(1.000)	(0.772)
T-4	-1.467	-0.164
	(0.961)	(0.820)
T-5	-1.418	-0.197
	(0.879)	(0.858)
T-6	-0.640	0.374
	(0.664)	(0.828)
Transfers	0.484***	0.476***
	(0.0326)	(0.0175)
N	5773	6052
Joint Pretrends Tests	0.2878	-
Cumulative Effect	-4.786	-2.293
	(11.974)	(8.344)
Likelihood Ratio Test	-	2.415
Bayes Test	-	0.241

Standard errors in parentheses
* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These are the results for Equation 2 controlling for the type of transfer used for that birth type.

Table 19: Results For Fresh Donor Births after the ACA controlling for Transfers

	(1) BJS FshDnrBirths	(3) OLS FshDnrBirths
T+0	0.536*	0.597*
	(0.320)	(0.332)
T+1	0.283	0.327
	(0.332)	(0.313)
T+2	0.437	0.603
	(0.414)	(0.372)
T+3	0.0890	0.296
	(0.454)	(0.419)
T+4	0.524	0.519
	(0.449)	(0.447)
T+5	0.330	0.209
	(0.480)	(0.445)
T+6	-0.0115	0.00589
	(0.471)	(0.448)
T-1	0.0284	
	(0.574)	
T-2	0.00907	0.0827
	(0.552)	(0.358)
T-3	0.143	0.126
	(0.546)	(0.359)
T-4	0.303	0.307
	(0.513)	(0.348)
T-5	-0.00640	0.0207
	(0.434)	(0.398)
T-6	-0.522	-0.528
	(0.375)	(0.385)
Transfers	0.547***	0.506***
	(0.0246)	(0.0182)
N	5773	6052
Joint Pretrends Tests	0.530	-
Cumulative Effect	2.187	2.555
	(2.151)	(2.107)
Likelihood Ratio Test	-	0.006
Bayes Test	-	0.249

Standard errors in parentheses
* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These are the results for Equation 2 controlling for the type of transfer used for that birth type.

Table 20: Results for Thawed Donor Transfers after SSM legalization

	(1) BJS ThwDnrTrans	(2) CS ThwDnrTrans	(3) OLS ThwDnrTrans
T+0	2.493	0.955	1.609
	(1.622)	(0.862)	(1.332)
T+1	7.900**	4.558**	6.853**
	(3.158)	(2.064)	(2.966)
T+2	7.175**	5.631*	5.731*
	(3.587)	(3.346)	(3.433)
T+3	4.201	2.353	3.118
	(3.043)	(3.335)	(3.026)
T-1	4.013		
	(2.551)		
T-2	1.953	-0.838	-0.857*
	(1.499)	(0.694)	(0.483)
T-3	0.675	-1.972	-0.993*
	(0.663)	(1.207)	(0.529)
N	5062		5129
Joint Pretrends Tests	0.3910	0.0157	-
Cumulative Effect	21.769**	13.497*	17.311**
	(8.482)	(7.492)	(8.349)
Likelihood Ratio Test	0	0	0.595
Bayes Test	0	0	0.218

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 1 with the outcome variable being the number of transfers performed by a clinic instead of the number of births.

Table 21: Results for Thawed Non-Donor Transfers after SSM legalization

	(1) BJS ThwNDTransfers	(2) CS ThwNDTransfers	(3) OLS ThwNDTransfers
T+0	5.548	0.432	-1.655
	(5.104)	(2.868)	(5.959)
T+1	18.17**	10.35*	11.23
	(9.015)	(6.122)	(10.18)
T+2	29.30***	17.75	17.14
	(10.07)	(11.14)	(14.48)
T+3	41.55***	30.06*	25.37
	(12.24)	(17.06)	(21.12)
T-1	21.35**		
	(8.548)		
T-2	12.91**	-2.904	-4.252*
	(5.621)	(2.917)	(2.436)
T-3	6.330**	-4.764	-4.197*
	(2.993)	(4.529)	(2.163)
N	5062		5129
Joint Pretrends Tests	0.2129	0.4774	-
Cumulative Effect	94.568***	58.591*	52.084
	(28.674)	(33.444)	(47.250)
Likelihood Ratio Test	0	0	1.963
Bayes Test	0	0	0.219

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 1 with the outcome variable being the number of transfers performed by a clinic instead of the number of births.

Table 22: Results for Fresh Donor Transfers after SSM legalization

	(1) BJS	(2) CS	(3) OLS
	FshDnrTransfers	FshDnrTransfers	FshDnrTransfers
T+0	-0.108	-2.202**	0.963
	(1.404)	(0.911)	(1.360)
T+1	-1.486	-5.813**	0.0145
	(1.992)	(2.660)	(2.148)
T+2	-1.236	-3.099	0.280
	(3.490)	(4.583)	(3.600)
T+3	-3.953	-5.884*	-1.854
	(2.660)	(3.216)	(2.557)
T-1	4.210		
	(3.497)		
T-2	1.765	-1.482*	-0.204
	(1.971)	(0.900)	(0.624)
T-3	0.677	-2.246	-0.540
	(0.938)	(1.401)	(0.628)
N	5062		5129
Joint Pretrends Tests	0.575	0.329	-
Cumulative Effect	-6.784	-16.998**	-0.597
	(7.262)	(9.051)	(8.110)
Likelihood Ratio Test	0	0	0.023
Bayes Test	0	0	0.218

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 1 with the outcome variable being the number of transfers performed by a clinic instead of the number of births.

Table 23: Results for Fresh Non-Donor Cycles after SSM legalization

	(1) BJS	(2) CS	(3) OLS
	FshNDCycle	FshNDCycle	FshNDCycle
T+0	-0.474	2.342	7.490
	(5.987)	(3.778)	(6.953)
T+1	-6.111	-10.28	2.488
	(8.875)	(7.126)	(9.735)
T+2	16.85	-10.25	14.17
	(18.10)	(16.98)	(14.55)
T+3	21.14	-7.088	23.94
	(19.94)	(21.35)	(18.09)
T-1	15.49		
	(18.88)		
T-2	20.21	10.21***	5.625
	(13.78)	(3.769)	(4.107)
T-3	13.27	7.928	0.762
	(9.054)	(5.524)	(4.110)
N	5062		5129
Joint Pretrends Tests	0.014	0.000	
Cumulative Effect	31.405	-25.274	48.091
	44.355	38.014	37.525
Likelihood Ratio Test	0	0	0
Bayes Test	0	0	0.216

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 1 with the outcome variable being the number of cycles performed by a clinic instead of the number of births.

Table 24: Results for Thawed Donor Transfers after the ACA

	(1) BJS	(3) OLS
	ThwDnrTrans	ThwDnrTrans
T+0	0.983	1.634
	(1.244)	(1.460)
T+1	2.398	3.635*
	(1.615)	(1.972)
T+2	5.342**	6.016**
	(2.487)	(2.771)
T+3	4.796	5.826*
	(3.188)	(3.419)
T+4	7.357**	7.711**
	(3.704)	(3.853)
T+5	9.888**	9.581**
	(4.318)	(4.351)
T+6	13.90**	12.67**
	(6.473)	(6.033)
T-1	2.310	
	(1.725)	
T-2	2.737*	1.966**
	(1.547)	(0.869)
T-3	1.743	0.726
	(1.561)	(0.928)
T-4	1.935	0.669
	(1.446)	(1.108)
T-5	1.641	0.504
	(1.006)	(0.925)
T-6	-0.215	-1.543
	(0.834)	(1.042)
N	5773	6052
Joint Pretrends Tests	0.265	-
Cumulative Effect	44.664**	47.069**
	(20.956)	(22.223)
Likelihood Ratio Test	0	0.000
Bayes Test	0	0.243

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 2 with the outcome variable being the number of transfers performed by a clinic instead of the number of births.

Table 25: Results for Thawed Non-Donor Transfers after the ACA

	(1) BJS	(3) OLS
	ThwNDTransfers	ThwNDTransfers
T+0	9.957***	9.405**
	(3.657)	(4.182)
T+1	12.76**	11.71*
	(5.570)	(6.396)
T+2	19.98**	21.19**
	(8.438)	(9.111)
T+3	22.35**	22.51**
	(9.732)	(10.86)
T+4	35.20***	34.43**
	(13.17)	(13.49)
T+5	56.98***	51.81***
	(17.63)	(16.99)
T+6	69.45***	62.59***
	(23.16)	(21.46)
T-1	13.29**	
	(6.131)	
T-2	6.673	1.195
	(4.917)	(3.324)
T-3	5.778	-0.908
	(4.104)	(3.514)
T-4	4.378	-3.266
	(3.539)	(3.713)
T-5	1.246	-6.757
	(2.880)	(4.150)
T-6	0.584	-7.925
	(2.080)	(4.816)
N	5773	6052
Joint Pretrends Tests	0.189	-
Cumulative Effect	226.673***	213.653***
	(76.064)	(77.828)
Likelihood Ratio Test	0	0.002
Bayes Test	0	0.232

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 2 with the outcome variable being the number of transfers performed by a clinic instead of the number of births.

Table 26: Results for Fresh Donor Transfers after the ACA

	(1) BJS	(3) OLS
	FshDnrTransfers	FshDnrTransfers
T+0	2.232	2.176
	(1.665)	(1.900)
T+1	2.683	2.343
	(2.319)	(2.603)
T+2	5.024	3.929
	(3.290)	(3.429)
T+3	5.106	4.100
	(3.157)	(3.326)
T+4	2.563	1.802
	(2.879)	(3.047)
T+5	-0.548	-0.375
	(2.963)	(3.091)
T+6	-2.609	-1.802
	(3.101)	(3.094)
T-1	3.968	
	(2.518)	
T-2	3.825	1.606
	(2.627)	(1.485)
T-3	2.449	0.343
	(1.874)	(1.085)
T-4	2.448	0.434
	(1.769)	(1.052)
T-5	2.208	0.517
	(1.586)	(0.988)
T-6	1.346	-0.00902
	(1.177)	(1.175)
N	5773	6052
Joint Pretrends Tests	0.881	-
Cumulative Effect	14.452	12.173
	(16.662)	(18.435)
Likelihood Ratio Test	0	0.005
Bayes Test	0	0.245

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 2 with the outcome variable being the number of transfers performed by a clinic instead of the number of births.

Table 27: Results for Fresh Non-Donor Cycles after the ACA

	(1) BJS FshNDCycle	(3) OLS FshNDCycle
T+0	11.86	6.561
	(9.875)	(9.736)
T+1	10.17	0.938
	(12.16)	(11.83)
T+2	-6.066	-12.97
	(13.57)	(12.73)
T+3	-4.310	-14.72
	(13.30)	(13.19)
T+4	-4.847	-13.40
	(13.99)	(13.90)
T+5	-16.58	-19.00
	(13.78)	(13.56)
T+6	-23.93	-23.94
	(16.07)	(15.14)
T-1	39.23**	
	(16.61)	
T-2	35.84**	13.67
	(16.39)	(8.519)
T-3	30.63*	11.06
	(16.87)	(9.925)
T-4	9.588	-8.462
	(10.85)	(6.349)
T-5	8.386	-8.444
	(7.970)	(6.547)
T-6	11.80**	-1.902
	(5.897)	(8.889)
N	5773	6052
Joint Pretrends Tests	0.017	-
Cumulative Effect	-33.693	-76.531
	(79.424)	(79.494)
Likelihood Ratio Test	0	0.001
Bayes Test	0	0.241

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 2 with the outcome variable being the number of transfers performed by a clinic instead of the number of births.

Table 28: Results For Thawed Donor Births after SSM using states with insurance mandates

	(1) BJS ThwDnrLvBirths	(2) CS ThwDnrLvBirths	(3) OLS ThwDnrLvBirths
T+0	2.193**	1.023*	1.070
	(1.014)	(0.580)	(0.805)
T+1	3.961***	2.878***	2.975**
	(1.526)	(1.030)	(1.264)
T+2	2.833**	2.512*	1.897*
	(1.194)	(1.447)	(0.983)
T+3	1.221	0.716	0.348
	(1.442)	(1.891)	(1.266)
T-1	1.680		
	(1.080)		
T-2	0.263	-0.901**	-0.842***
	(0.651)	(0.430)	(0.319)
T-3	0.343	-1.215	-0.370
	(0.360)	(0.759)	(0.299)
N	2608		2688
Joint Pretrends Tests	0.192	0.735	-
Cumulative Effect	10.208***	7.129*	6.290**
	(3.814)	(3.936)	(3.148)
Likelihood Ratio Test	0	0	32.154
Bayes Test	0	0	0.221

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 1 without transfers and is limited to only the states that had insurance mandates before the ACA. These results are also the tabular results for Figure 4.

Table 29: Results For Thawed Non-Donor Births after SSM using states with insurance mandates

	(1) BJS ThwNDLvBirths	(2) CS ThwNDLvBirths	(3) OLS ThwNDLvBirths
T+0	2.525	0.793	-2.942
	(3.964)	(1.835)	(3.714)
T+1	3.760	2.242	-0.424
	(6.222)	(4.194)	(5.446)
T+2	10.19**	8.542	3.241
	(5.046)	(5.971)	(6.408)
T+3	10.05	5.912	2.078
	(7.364)	(10.29)	(8.940)
T-1	9.029**		
	(4.577)		
T-2	5.903**	-1.199	-1.376
	(2.588)	(2.710)	(2.102)
T-3	3.264***	-1.421	-1.086
	(1.259)	(0.914)	(1.407)
N	2608		2688
Joint Pretrends Tests	0.300	0.009	-
Cumulative Effect	26.524	17.490	1.953
	(19.396)	(20.656)	(21.916)
Likelihood Ratio Test	0	0	0.189
Bayes Test	0	0	0.219

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 1 without transfers and is limited to only the states that had insurance mandates before the ACA. These results are also the tabular results for Figure 4.

Table 30: Results For Fresh Donor Births after SSM using states with insurance mandates

	(1) BJS FshDnrBirths	(2) CS FshDnrBirths	(3) OLS FshDnrBirths
T+0	0.430	-1.490*	0.827
	(0.974)	(0.856)	(0.884)
T+1	-0.588	-2.928*	-0.0181
	(1.071)	(1.671)	(1.264)
T+2	0.453	0.516	0.988
	(1.512)	(1.835)	(1.457)
T+3	-2.123	-1.767	-0.962
	(1.449)	(1.491)	(1.220)
T-1	2.157		
	(1.812)		
T-2	0.552	-1.421	0.0725
	(1.064)	(0.914)	(0.707)
T-3	0.282	-1.990*	-0.265
	(0.701)	(1.186)	(0.549)
N	2608		2688
Joint Pretrends Tests	0.5098	0.2653	-
Cumulative Effect	-1.829	-5.669	0.836
	(3.573)	(4.638)	(3.833)
Likelihood Ratio Test	0	0	0.020
Bayes Test	0	0	0.219

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 1 without transfers and is limited to only the states that had insurance mandates before the ACA. These results are also the tabular results for Figure 4.

Table 31: Results For Fresh Non-Donor Births after SSM using states with insurance mandates

	(1) BJS FshNDLvBirths	(2) CS FshNDLvBirths	(3) OLS FshNDLvBirths
T+0	2.055	0.952	2.386
	(2.832)	(2.181)	(2.739)
T+1	0.130	0.228	0.381
	(3.352)	(3.737)	(3.004)
T+2	3.369	1.033	0.692
	(6.254)	(7.975)	(5.680)
T+3	-1.263	-2.187	-0.966
	(8.308)	(10.87)	(6.915)
T-1	0.493		
	(5.423)		
T-2	1.307	1.731	0.736
	(3.914)	(2.225)	(1.916)
T-3	0.277	0.625	-0.258
	(2.866)	(3.175)	(1.720)
N	2608		2688
Joint Pretrends Tests	0.7525	0.1378	-
Cumulative Effect	4.291	0.026	2.492
	(17.881)	(22.172)	(15.014)
Likelihood Ratio Test	0	0	0.008
Bayes Test	0	0	0.219

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results use Equation 1 without transfers and is limited to only the states that had insurance mandates before the ACA. These results are also the tabular results for Figure 4.

Table 32: Results for Number of Clinics in a City after the ACA

	(1) BJS howmanycity	(2) OLS howmanycity
T+0	0.183	0.0941
	(0.149)	(0.142)
T+1	0.317	0.201
	(0.200)	(0.201)
T+2	0.287*	0.170
	(0.150)	(0.138)
T+3	0.337**	0.207
	(0.149)	(0.142)
T+4	0.240*	0.104
	(0.136)	(0.119)
T+5	0.287*	0.146
	(0.162)	(0.151)
T+6	0.265	0.118
	(0.167)	(0.162)
T-1	0.399*	
	(0.241)	
T-2	0.140	-0.0778
	(0.155)	(0.105)
T-3	0.0107	-0.191
	(0.132)	(0.173)
T-4	0.171	-0.0279
	(0.169)	(0.0877)
T-5	0.153	-0.0441
	(0.132)	(0.125)
T-6	0.125	-0.0687
	(0.0808)	(0.0823)
N	5947	6107
Joint Pretrends Tests	0.4336	-
Cumulative Effect	1.9165*	1.0391
	(0.9966)	(0.9366)
Likelihood Ratio Test	-	0.1348
Bayes Test	-	0.2467

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Note: These results display the results from Equation 3 for the ACA. These results are also the tabular results for Figure 5.

Table 33: Results for Number of Clinics in a City after SSM Legalization

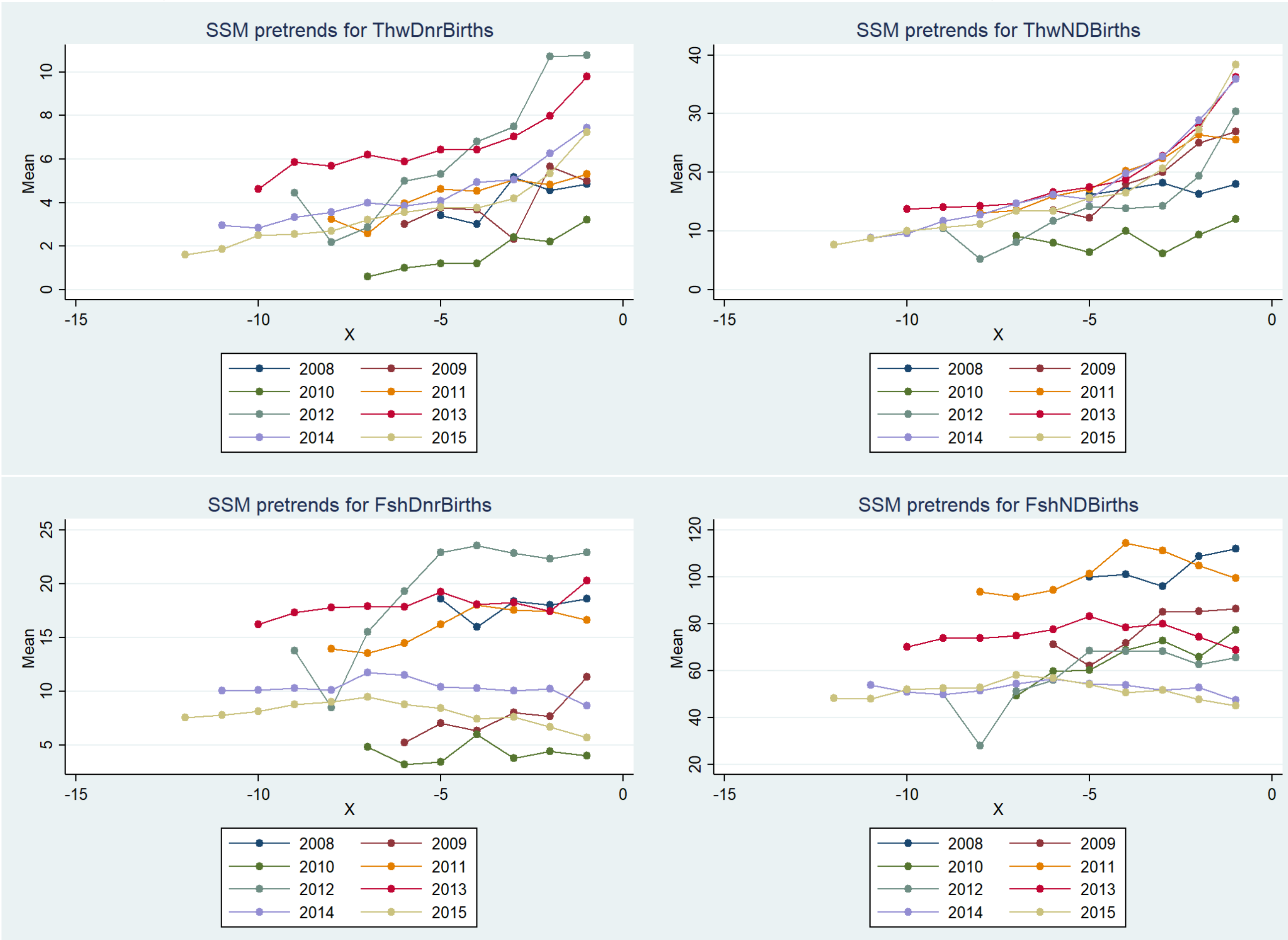
	(1) BJS howmanycity	(2) OLS howmanycity
T+0	0.0956	0.155
	(0.0898)	(0.192)
T+1	0.120	0.185
	(0.110)	(0.190)
T+2	0.373***	0.323
	(0.125)	(0.269)
T+3	0.0162	0.0273
	(0.101)	(0.163)
T-1	0.444	
	(0.375)	
T-2	0.318	-0.00755
	(0.283)	(0.0475)
T-3	0.164	-0.0772
	(0.121)	(0.0787)
N	5089	5183
Joint Pretrends Tests	0.2480	-
Cumulative Effect	0.6043	0.6912
	(0.3751)	(0.7916)
Likelihood Ratio Test	-	0.0716
Bayes Test	-	0.2208

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

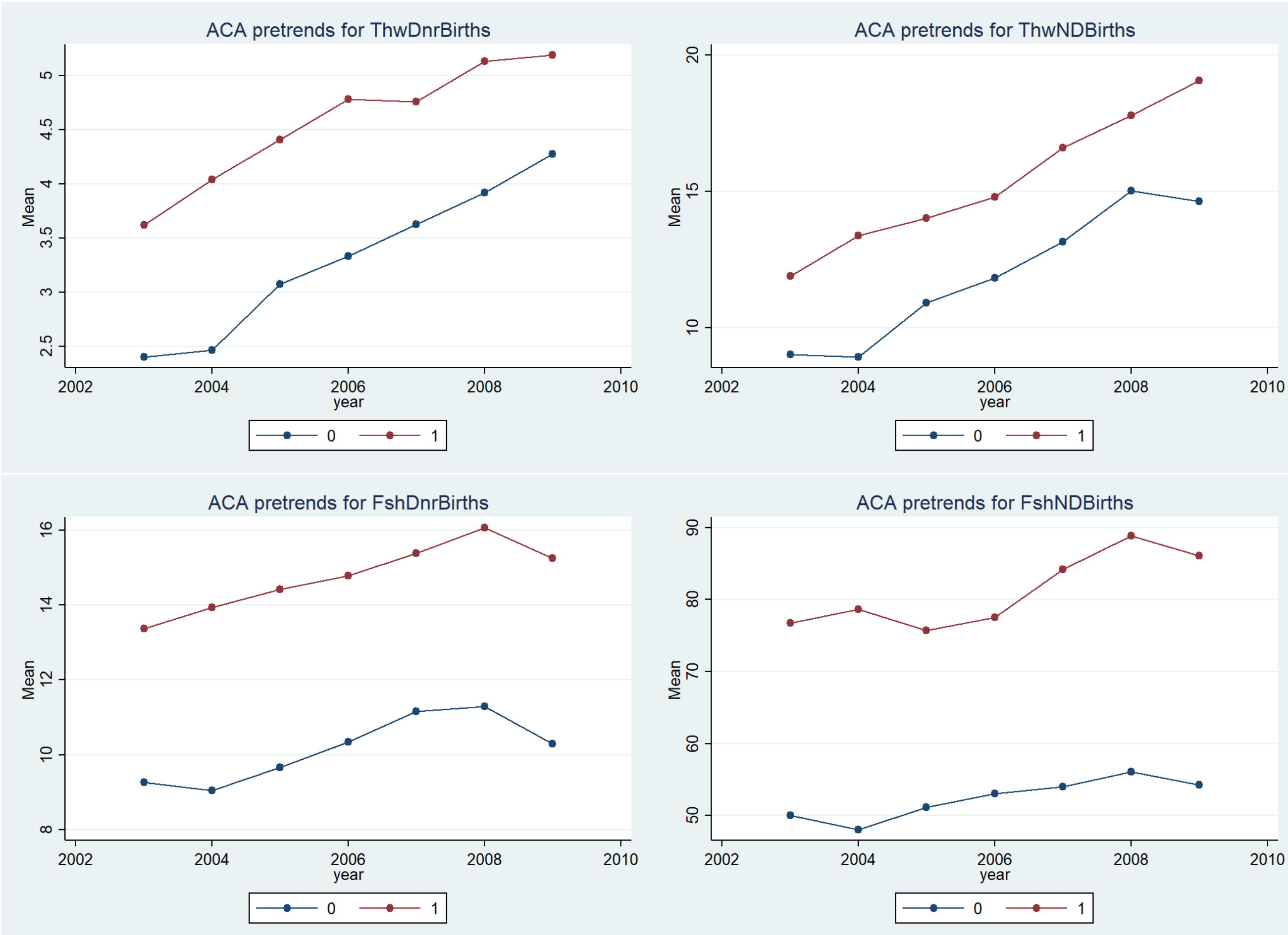
Note: These results display the results from Equation 3 for SSM legalization. These results are also the tabular results for Figure 5.

Figure 7: Trends for the different types of births before SSM legalization



Note: These charts show the trends by year of legalization before SSM legalization. These show that there are very similar pretrends across the four birth types.

Figure 8: Trends for the different types of births before the ACA



Note: These charts show the trends by whether they have insurance mandates in place or not before passage of the ACA. These show that there are very similar pretrends across the four birth types.